THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The 12th term of the AP whose 1st term is 7 and 5th term is 19 is: 1

(a) 39 **(b)** 42 **(c)** 40 **(d)** 41

2. If $DE \parallel BC$ in $\triangle ABC$, $AB = 5$ cm, $AD = 3$ cm, $AC = 10$ cm, then EC (cm) is: 1

(a) 4 **(b)** 5 **(c)** 6 **(d)** 3

3. The perimeter of the triangle with vertices $(0,0)$, $(6,0)$, $(0,8)$ is: 1

(a) 26 **(b)** 24 **(c)** 23 **(d)** 25

4. The number of terms in the AP 5, 9, \dots, 97 is: 1

(a) 26 **(b)** 25 **(c)** 24 **(d)** 23

5. The areas of two similar triangles are 36 cm² and 100 cm². The ratio of their perimeters is 3: 1

(a) 5 **(b)** 7 **(c)** 6 **(d)** 4

6. The point which divides $(6,0)$ and $(0,10)$ in the ratio 1:1 has coordinates summing to: 1

(a) 10 **(b)** 9 **(c)** 7 **(d)** 8

7. The sum of all two-digit multiples of 3 is: 1

(a) 1664 **(b)** 1667 **(c)** 1666 **(d)** 1665

8. The altitude to the hypotenuse of a right triangle divides it into 4 cm and 9 cm. The altitude (cm) is: 1

(a) 6 **(b)** 8 **(c)** 7 **(d)** 5

9. The centroid of the triangle with vertices $(1,2)$, $(4,6)$, $(7,1)$ has x-coordinate: 1

(a) 5 **(b)** 6 **(c)** 4 **(d)** 3

10. The sum of the first 14 terms of the AP $12, 17, 22, 27, \ldots$ is: 1

(a) 624 **(b)** 622 **(c)** 623 **(d)** 625

11. A pole 6 m high casts a shadow 4 m long. A tower casting a shadow 24 m has height (m): 1

(a) 37 **(b)** 36 **(c)** 35 **(d)** 38

12. A circle has centre $(1,2)$ and passes through $(4,6)$. Its radius is: 1

(a) 7 **(b)** 5 **(c)** 6 **(d)** 4

13. The middle term of an AP with 13 terms, first term 5 and last term 41, is: 1

(a) 22 **(b)** 24 **(c)** 25 **(d)** 23

14. The diagonals of a rhombus are 12 and 16. Its side squared is: 1

(a) 102 **(b)** 100 **(c)** 99 **(d)** 101

15. The ratio $k : 1$ in which the line $x = 5$ divides the join of $(2, 0)$ and $(11, 0)$, with $a < 5 < c$, has $k =$ 1

(a) $\frac{1}{2}$ **(b)** $\frac{5}{2}$ **(c)** $-\frac{1}{2}$ **(d)** $\frac{3}{2}$

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Which term of the AP $7, 13, 19, 25, \ldots$ is 127? 2

17. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 5$ cm, $AD = 3$ cm and $AC = 10$ cm, find AE and EC. 2

18. Find the values of y for which the distance between $P(7,\ y)$ and $Q(1,\ 2)$ is 10 units. 2

19. Find the sum of the first 18 terms of the AP $14, 20, 26, 32, \ldots$ 2

20. Two similar triangles have perimeters 21 cm and 42 cm. If a side of the smaller is 4 cm, find the corresponding side of the larger and the ratio of their areas. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. In an AP, the 3rd term is 19 and the 8th term is 54. Find the AP and its 15th term. 3

22. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 9$ cm, $AD = 4$ cm and $AC = 18$ cm, find AE and EC. 3

23. Show that the points $(0,\ 0)$, $(4,\ 0)$ and $(2,\ 3)$ are the vertices of an isosceles triangle. 3

24. Find the sum of the first 20 terms of the AP whose nth term is $a_n = 3n + 2$. 3

OR

$\triangle ABC \sim \triangle PQR$ with $AB = 3$ cm, $BC = 4$ cm, $CA = 5$ cm and $PQ = 6$ cm. Find QR and RP. 3

25. Find the coordinates of the point which divides the join of $(-2,\ 3)$ and $(8,\ 13)$ internally in the ratio $2 : 3$. 3

OR

The sum of the first 10 terms of an AP is 185 and the first term is 5. Find d and the 10th term. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. How many terms of the AP $54, 50, 46, 42, \ldots$ must be taken so that the sum is 360? 5

OR

Two poles of heights 14 m and 2 m stand on level ground 9 m apart. Find the distance between their tops. 5

27. In what ratio does the x-axis divide the segment joining $(-3,\ -5)$ and $(6,\ 4)$? Find the point of division. 5

OR

A ladder has 11 rungs whose lengths decrease uniformly from 45 cm at the bottom to 25 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. 5

END OF QUESTION PAPER **Total: 50 Marks**

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